

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

Assignment 2

CE 308/408 Cloud Computing

Cloud Infrastructure for StreamNest Analytics using Google Cloud Platform (GCP)

Introduction

StreamNest is a fast-growing video streaming startup that has recently expanded its user base from 50,000 to over 2 million active users across Southeast Asia. The company collects massive amounts of data daily including user watch history, clickstream events, content metadata, video quality metrics, and subscription billing records. Currently, all of this data is stored in a mix of on-premises MySQL databases and unorganized flat files on local servers, making it nearly impossible for the data and engineering teams to collaborate or derive meaningful insights in a timely manner.

The CTO of StreamNest has decided to migrate the entire infrastructure to **Google Cloud Platform (GCP)** and has hired your team as cloud architects to design and implement a modern, scalable solution.

Problem:

StreamNest has two core problems that need to be solved simultaneously. First, the engineering team has built its backend as a set of independent microservices including a user authentication service, a content recommendation engine, and a billing service — that currently run on bare virtual machines and suffer from environment inconsistencies between development and production. Deployments are manual, error-prone, and take hours to complete. Second, the data science team has no unified platform to access clean, structured data for building recommendation models or generating business reports. Raw event logs arrive in different formats some in real-time bursts during peak hours, others in nightly batch dumps with no organized system to bring them together.

Requirement:

You are expected to produce a design document along with a working proof-of-concept on GCP using your available credits. The proof-of-concept does not need to replicate the entire StreamNest platform it needs to demonstrate that your proposed architecture works end-to-end.

For the containerization side, you should containerize at least one microservice that represents StreamNest's backend such as a simple content catalog API or a user activity logger that accepts and returns JSON data. This service should be deployed and running on **Google Kubernetes Engine (GKE)**, accessible via a public endpoint, and demonstrate that it can be scaled and redeployed without downtime.

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For the data side, you should construct a working data lakehouse pipeline on GCP using a realistic sample dataset that resembles what StreamNest would produce simulated user watch events containing fields like user ID, content ID, watch duration, device type, timestamp, and region. This raw dataset should land in **Google Cloud Storage** as the raw ingestion layer, be made queryable through **BigQuery** as an external table, and result in at least one clean aggregated table that answers a real business question such as which content received the most watch time, or which device type is most popular in a given region.

Deliverable

You are required to submit a written design document that explains your architecture and includes screenshots of your working implementation embedded within it. The document should be written as a professional proposal to StreamNest's CTO explaining what you built, why you made the design decisions you did, and what your proof-of-concept demonstrates about the scalability of the approach. Screenshots should show your running GKE deployment, the public API response, your Cloud Storage bucket with the raw dataset, and your final BigQuery analytics table with results.